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Paper 1: Anti-PD-(L)1 Antibodies: Insights From QSP-Based Meta-Analysis

Bibliographic Information

- **Title**: Anti-PD-(L)1 Antibodies: Insights From QSP-Based Meta-Analysis
- **Authors**: Carter L. Johnson, et al.
- **Journal**: CPT: Pharmacometrics & Systems Pharmacology, 2026
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Quick Take

A mechanistic QSP analysis challenging the dogma that clinical efficacy differences between anti-PD-1 and anti-PD-L1 antibodies are driven by target occupancy. The study demonstrates that while clinical meta-analyses favor anti-PD-1, QSP simulations show anti-PD-L1s actually achieve superior complex inhibition, suggesting alternative mechanisms (e.g., PD-L2) drive the clinical divergence.

Executive Summary

Clinical meta-analyses have consistently suggested that anti-PD-1 monoclonal antibodies (mAbs) yield better survival outcomes than anti-PD-L1 mAbs, despite a shared hypothesis that both classes work by inhibiting the PD-1:PD-L1 signaling axis. This study utilizes a Quantitative Systems Pharmacology (QSP) model to mechanistically test whether differential inhibition of the PD-1:PD-L1 complex explains this efficacy gap. By simulating eight approved mAbs (5 anti-PD-1, 3 anti-PD-L1) at clinical doses across varying tumor burdens, the authors found that all agents achieve >96% complex inhibition. Contrary to clinical efficacy trends, the model predicts that anti-PD-L1 antibodies actually sustain slightly higher levels of complex inhibition than anti-PD-1s. These findings imply that the 'shared mechanism' hypothesis is incomplete; the superior clinical performance of anti-PD-1s cannot be attributed to better blockade of the PD-1:PD-L1 interaction, pointing instead to unmodeled pathways such as PD-L2 or B7.2 interactions.

⚡ Methodological Snapshot

The authors employed a 'Monospecific Anti-Receptor with Membrane Ligand Competitor' QSP model within the Applied BioMath QSP Notebook framework. The model features a four-compartment structure (Central, Peripheral, Disease/Tumor) describing the PK/PD of mAbs binding to membrane-bound receptors (PD-1) or ligands (PD-L1).

Simulations were conducted for eight approved mAbs (pembrolizumab, nivolumab, cemiplimab, dostarlimab, retifanlimab, atezolizumab, avelumab, durvalumab) using literature-derived binding affinities and PK parameters. The analysis included six tumor parameterization scenarios varying PD-L1 expression (30-90%+) and Effector:Target (E:T) ratios (1:10 to 1:100). A virtual population

(Vpop) approach was used for sensitivity analysis and to perform a bootstrap comparison mirroring the clinical meta-analysis by Duan et al.

🏗 Structural Model Breakdown

The model consists of Central, Peripheral, and Disease (Tumor) compartments. It describes the kinetics of a mAb distributing to the tumor and binding its target (PD-1 on T-cells or PD-L1 on tumor cells). Key processes include: 1. Linear elimination and distribution between central/peripheral compartments. 2. Reversible binding of mAb to target. 3. Formation of the PD-1:PD-L1 trans-complex between T-cells and tumor cells. 4. Internalization of free receptors, drug-receptor complexes, and PD-1:PD-L1 complexes. 5. Target-mediated drug disposition (TMDD) is emergent based on receptor density and turnover.

Detailed Methodological Analysis

Modeling Approach

Mechanistic QSP model (4-compartment) incorporating TMDD, receptor internalization, and competitive binding kinetics. Implemented in Applied BioMath QSP Notebook.

Data Sources

Parameters (affinity, half-life, receptor density) derived from published literature and meta-analyses. Clinical dosing regimens for 8 approved mAbs were simulated.

Estimation Methods

Simulation-based study. No parameter estimation from patient data was performed in this specific work; parameters were fixed based on literature values.

Model Evaluation

Sensitivity analysis using Virtual Patients (Vpop) to assess impact of parameter variability. Bootstrap sampling (10,000 samples) used to statistically compare predicted inhibition levels between drug classes.

Covariate Analysis

Explored impact of tumor burden, PD-L1 expression levels, and T-cell density (E:T ratio) as structural scenarios rather than statistical covariates.

Scientific Context & Motivation

The field of immuno-oncology lacks direct head-to-head clinical trials comparing anti-PD-1 and anti-PD-L1 agents, relying instead on meta-analyses which suggest a survival advantage for anti-PD-1s. The prevailing assumption has been that efficacy is driven solely by the disruption of PD-1:PD-L1 binding. This study addresses a critical knowledge gap by using mechanistic modeling to isolate the variable of 'complex inhibition.' It challenges the paradigm that target occupancy

is the sole driver of efficacy differentiation and provides a biophysical basis for investigating secondary signaling pathways (PD-L2, B7.2) as the true differentiators.

Statistical Rigor Assessment

The study employs a robust Vpop approach (500 virtual patients) to account for inter-individual variability in physiological and system parameters. The bootstrap analysis (10k iterations) specifically mirrored the design of a pivotal clinical meta-analysis (Duan et al.), providing a rigorous statistical basis for the conclusion that complex inhibition differences do not align with clinical survival hazard ratios.

📊 Key Findings

1. **Target Engagement:** All 8 mAbs achieve >97% target engagement at clinical steady state.
2. **Complex Inhibition:** All mAbs achieve >96% inhibition of PD-1:PD-L1 complex formation. Durvalumab (anti-PD-L1) showed the highest inhibition (~99.99%).
3. **Class Comparison:** Anti-PD-L1 mAbs consistently demonstrated equal or slightly higher complex inhibition compared to anti-PD-1 mAbs, directly contradicting the clinical trend where anti-PD-1s show superior efficacy.
4. **Bootstrap Result:** In a direct comparison of Pembrolizumab vs. Avelumab (mirroring the Duan et al. meta-analysis), only 3.92% of bootstrap samples showed Pembrolizumab having higher complex inhibition, whereas clinically Pembrolizumab has a significant survival advantage (HR 0.73).

💡 Clinical & Regulatory Implications

The results suggest that increasing doses to achieve higher PD-1:PD-L1 complex inhibition is unlikely to yield clinical benefit, as inhibition is already saturated. For drug developers and regulators, this implies that 'complex inhibition' is a poor surrogate endpoint for comparing efficacy between these two classes. Future development should focus on the differential effects of blocking PD-L2 (unique to anti-PD-1s) or B7.2 interactions (unique to anti-PD-L1s).

Strengths & Limitations

Strengths

- Comprehensive inclusion of all 8 approved mAbs with drug-specific parameters.
- Direct mechanistic testing of a clinical hypothesis (occupancy vs. efficacy).
- Use of Vpop and bootstrap methods to rigorously assess variability and statistical significance.
- Clear separation of binding kinetics from downstream efficacy assumptions.

Limitations (Acknowledged by Authors)

- Assumption of constant tumor cell number (no efficacy feedback loop on tumor size).
- Reliance on literature-derived parameters which may have inherent uncertainty.
- Exclusion of other pathways (PD-L2, B7.2) in the current model structure.

Limitations (Expert Review)

- The model predicts occupancy/inhibition but does not translate this to T-cell killing rates or tumor shrinkage, limiting the ability to simulate actual survival curves.
- Potential differences in tissue penetration (tumor accessibility) between different antibodies were not explicitly detailed as a variable.

Generalizability

Highly generalizable for understanding the biophysics of checkpoint blockade, but specific clinical outcomes may depend on tumor-specific immune microenvironments not fully captured in the generic tumor compartment.

Figures & Tables

- **Figure 1**: Diagram of the 4-compartment QSP model showing Central, Peripheral, and Disease compartments, with receptor-ligand binding interactions.
  - *Significance*: Visualizes the structural framework and the specific mechanistic interactions (binding, internalization, competition) being simulated.
- **Figure 2**: Simulation results for PK, target engagement, and complex inhibition for all 8 mAbs across different tumor burden scenarios.
  - *Significance*: Demonstrates that all agents achieve near-complete target saturation (>96% inhibition) at clinical doses, with anti-PD-L1s showing slightly higher inhibition.
- **Figure 3**: Virtual Population (Vpop) simulations showing variability in complex inhibition.
  - *Significance*: Confirms that even with inter-individual variability, the conclusion holds: anti-PD-L1s consistently achieve high inhibition levels comparable to or better than anti-PD-1s.
- **Figure 4**: Bootstrap analysis comparing Pembrolizumab vs. Avelumab inhibition.
  - *Significance*: Statistically demonstrates the disconnect: while Pembrolizumab is clinically superior, the model shows it has lower/equal complex inhibition in ~96% of virtual cases.

Code & Reproducibility Assessment

The model was implemented in 'Applied BioMath QSP Notebook' (Certara), which appears to be proprietary commercial software. While parameters are provided in supplementary materials, the code itself is likely not open-source.

Supplementary Materials

Supplementary materials include Table S1 (complete model parameters derived from literature), Appendix S1 (details on Vpop generation and tumor parameterization scenarios), and Figure S1 (dosing alignment). These provide the necessary inputs to reproduce the simulations.

Future Directions

1. Expansion of the QSP model to include PD-1:PD-L2 and PD-L1:B7.2 signaling axes to test if these explain the efficacy gap.

2. Integration of this mechanistic framework with Model-Based Meta-Analysis (MBMA) to identify relevant covariates.
3. Investigation of resistance mechanisms such as PD-L1 upregulation or neoantigen expression.

Expert Commentary

This is a high-value 'negative result' paper. By definitively showing that PD-1:PD-L1 complex inhibition is effectively saturated by both drug classes and cannot explain the clinical efficacy gap, the authors force the field to look elsewhere. It is a textbook example of using QSP to prune hypotheses. The finding that anti-PD-L1s are theoretically *better* at inhibition (likely due to target abundance and kinetics) yet clinically *worse* is a compelling paradox that highlights the importance of the PD-L2 axis, which anti-PD-L1s leave intact. This paper should be required reading for anyone designing next-generation checkpoint inhibitors.

Bottom Line

Practicing pharmacometricians should recognize that for PD-1/PD-L1 inhibitors, 'target occupancy' and 'complex inhibition' are saturated at clinical doses and are poor differentiators of efficacy. The clinical superiority of anti-PD-1s over anti-PD-L1s is not a PK or binding phenomenon, but likely a systems biology phenomenon involving unblocked pathways like PD-L2. QSP models must incorporate these secondary axes to accurately predict comparative efficacy.

📊 Figures

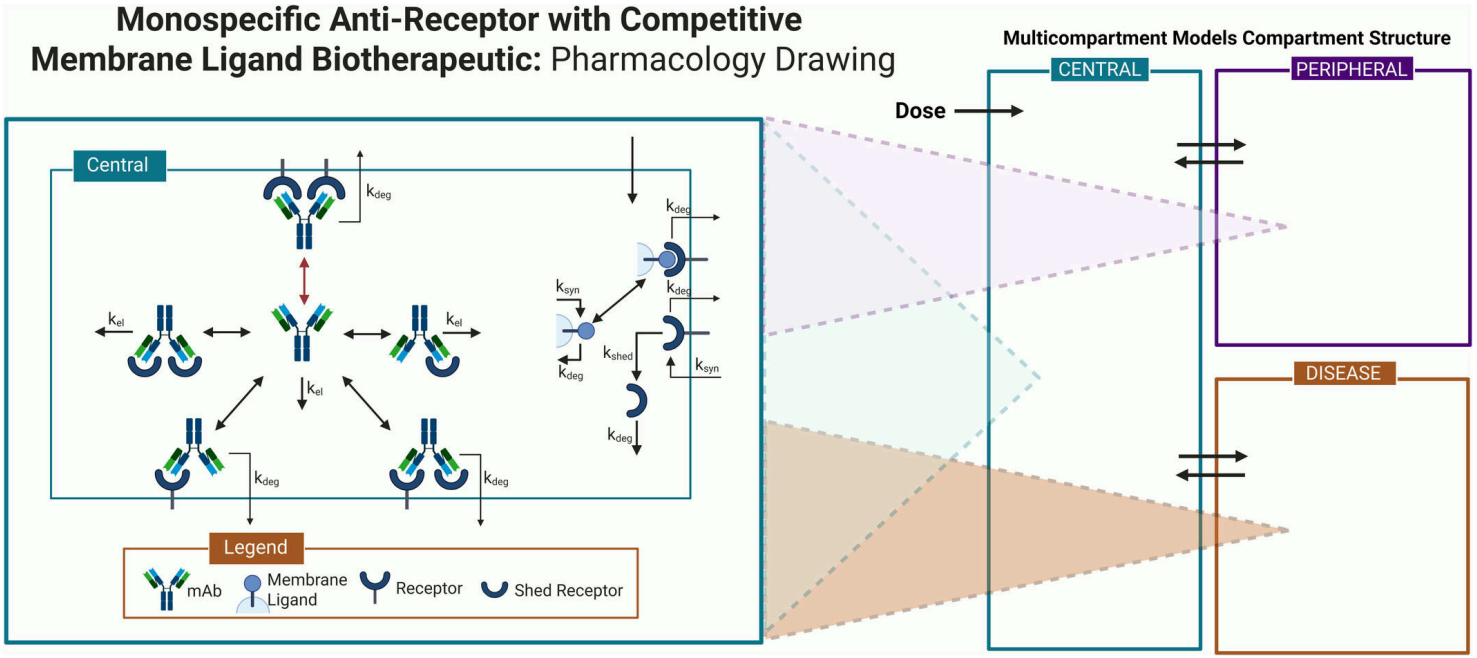

Figure 1: FIGURE 1: Model diagram. After administration in the Central compartment, the antibody can distribute to the Peripheral and Disease (Tumor) compartments. Elimination occur

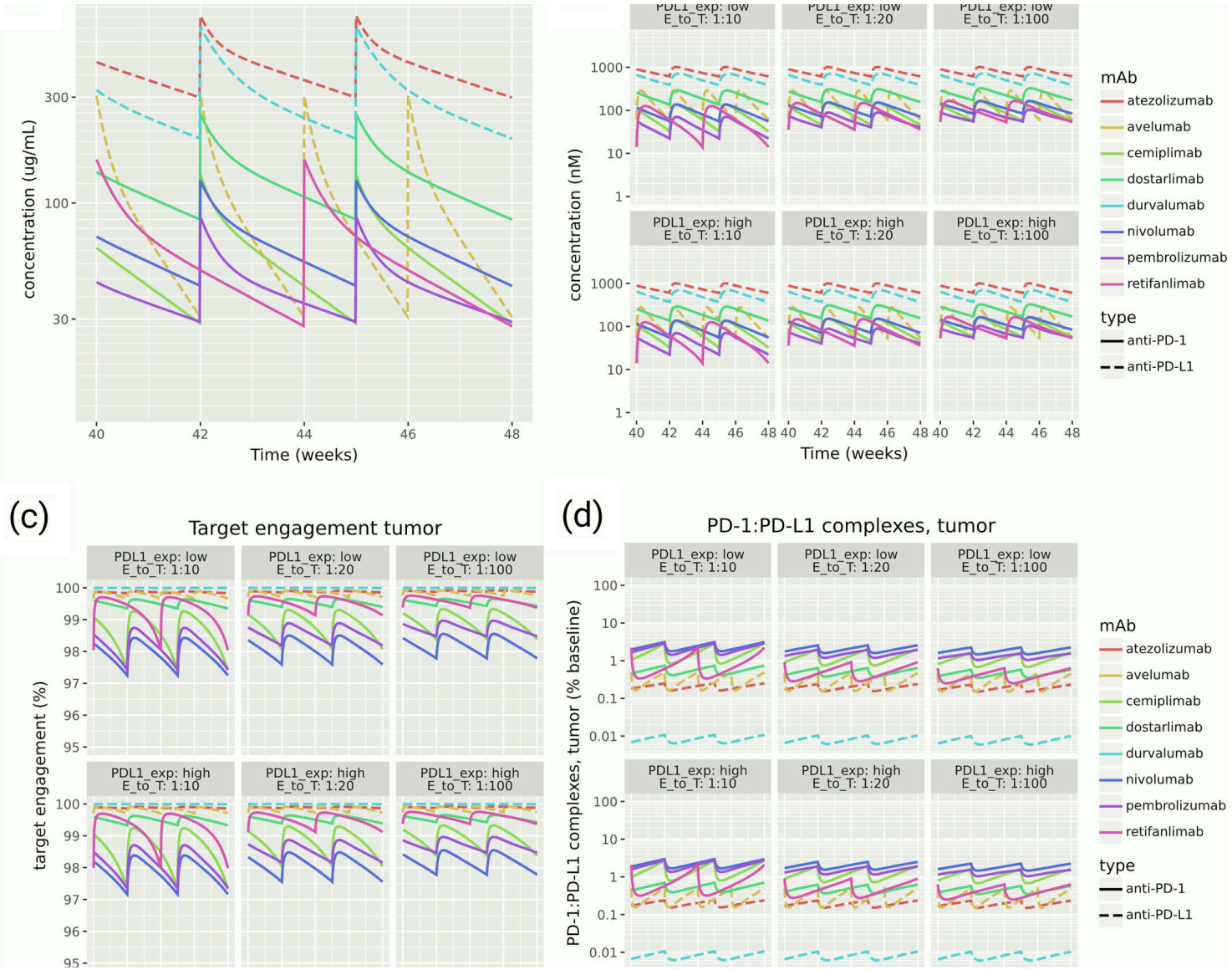

Figure 2: FIGURE 2: Model simulations of the eight mAbs at their respective clinical doses. (a) Steady state plasma PK profiles for the eight mAbs. (b) Steady state tumor PK profile

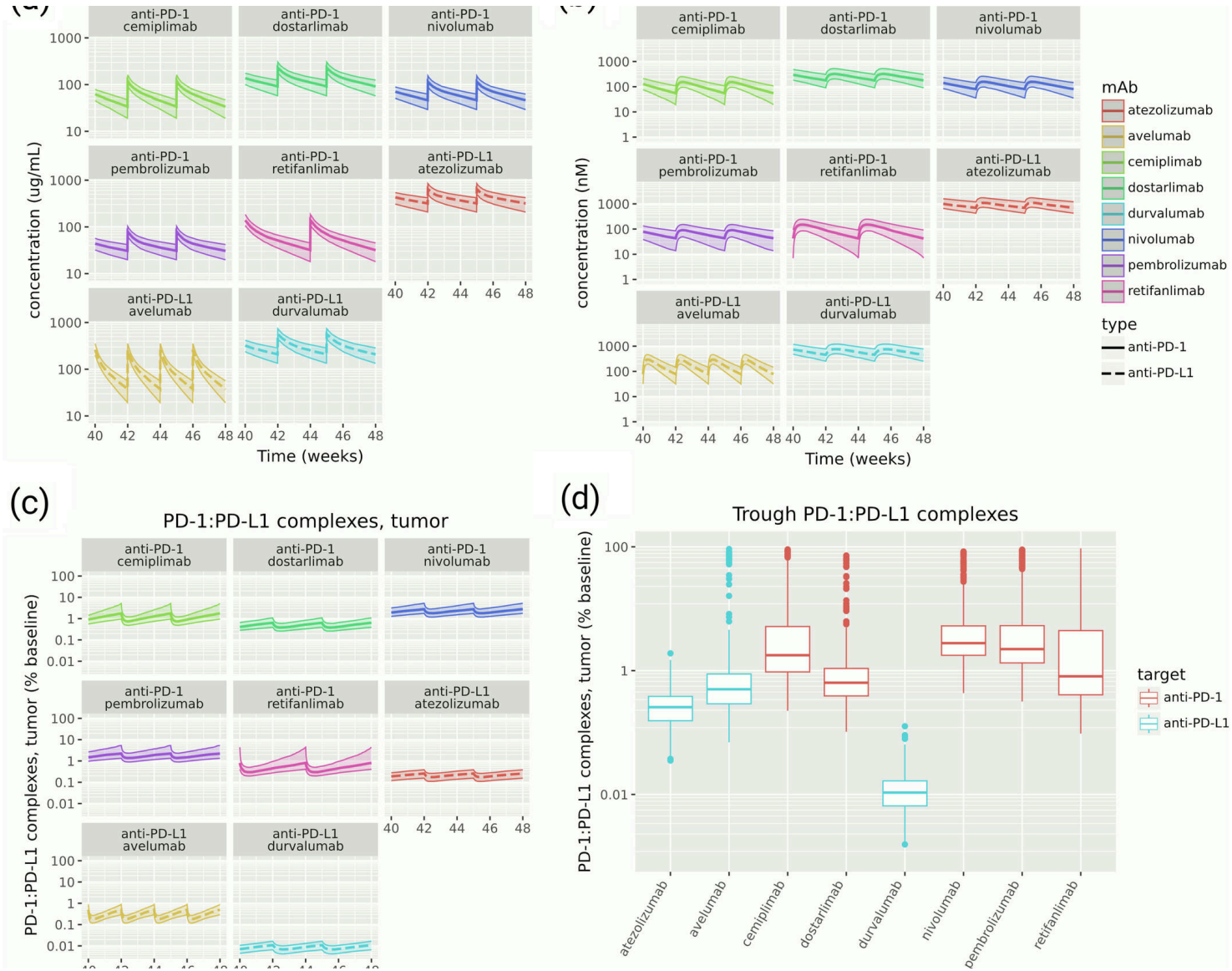

Figure 3: FIGURE 3: Vpop simulations of the eight mAbs at their respective clinical doses. Ribbons indicate the lower and upper quartile profiles, with the central solid/dashed line

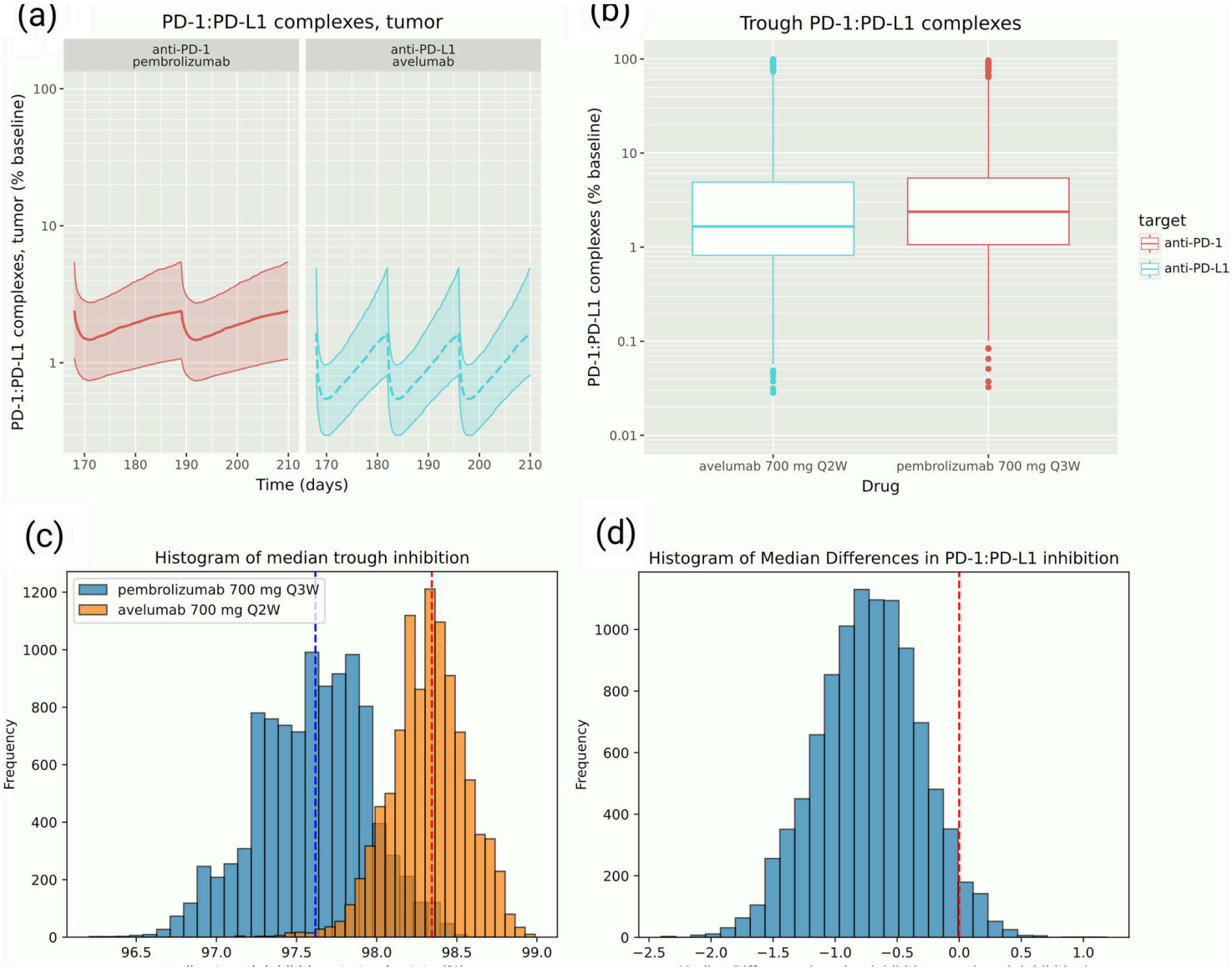

Figure 4: FIGURE 4: Bootstrap analysis of avelumab 10 mg/kg Q2W vs. pembrolizumab 10 mg/kg Q3W. (a) Vpop simulation of PD-1:PD-L1 complex inhibition in the tumor (% baseline) for th